

EE201: Circuit Theory I

Fall 2026–2027 | Course Syllabus

1. Instructor Information

Sec.	Instructor	Office	e-mail	Sec.	Instructor
1	T. Engin Tuncer	E109	etuncer@metu.edu.tr	4	Emre Tuna (E113)
2	Mert Ankaralı	A-410/D-104	mertan@metu.edu.tr	5	Yunus Aslan (C-103)
3	Emre Tuna	E113	etuna@metu.edu.tr		

Teaching assistants — **EE201 (theory)**: Anıl Gülbeden (gulbeden@metu.edu.tr).

EE201/EE213 — laboratory: Alp Gürsoy, Eda Özkaynar, Orhun Koç, Civan Serhat Çevik, Elif Çağla Tay, Enfal Çabuk, Adem Utku Atasayar, Özgür Gülsuna, Alper Alka.

2. Textbooks, References and Other Resources

- Alexander, C. K., & Sadiku, M. N. O. (2016). *Fundamentals of Electric Circuits*. McGraw-Hill. (main textbook)
- Chua, L. O., Desoer, C. A., & Kuh, E. S. (1987). *Linear and Nonlinear Circuits*. McGraw-Hill. (reference)
- Lecture notes and videos by Emre Tuna, users.metu.edu.tr/etuna/ee201.

ODTÜCLASS (odtuclass.metu.edu.tr) is the official channel for all sections: announcements, the detailed course outline, grades, exam and laboratory information, and the common course material listed above.

3. Course Materials for Section 2

In addition to the common resources above, the lecture notes, in-class examples, extra problems and supplementary material *specific to Section 2* are maintained in a public repository:

Section 2 lecture notes and course material

github.com/mertankarali/Lecture-Notes → METU-EE201

Public: no account is required to read or download. Updated *during* the semester, typically before each lecture — refresh it regularly rather than downloading once. Corrections and typo reports (issues or e-mail) are very welcome.

What is where: *repository* → Section 2 lecture notes, worked examples, extra problems and solutions, code and simulation files; *ODTÜCLASS* → everything official and everything common to all sections. Announcements are always made on ODTÜCLASS.

4. Grading

Component	Attendance	Midterm 1	Midterm 2	Final Exam	Laboratory	Total
Weight	5 %	20 %	20 %	35 %	20 %	100 %

5. Attendance (Section 2)

Attendance is worth 5 % of the grade, and attendance below 40 % means **NA** (see Section 6). In Section 2 it is collected as follows.

- Attendance is taken with a **QR code through ODTÜCLASS**. Come with a device that can read it and with your ODTÜCLASS session ready.
- It is taken **once per class meeting** — once in a one-hour meeting and once in a two-hour meeting, *not* once per lecture hour.
- It is normally opened **at the beginning** of the meeting. In a two-hour meeting, if I give a break, I may open it **at the break** instead.
- Meetings are not always run the same way: I sometimes teach two hours as a single block, sometimes finish early, and I do not always open attendance at the break.
- Therefore there is **no pattern to rely on**. Missing the first hour and arriving at the break — or any similar strategy — is entirely at your own risk: if attendance was taken while you were not in the room, that meeting is lost and it is not repeated.

6. EE201 Final Examination Policy

A student (i) whose attendance is below 40 %, (ii) missing any midterm examination without a valid excuse, (iii) missing any laboratory experiment without a valid excuse, or (iv) missing the Laboratory Final Exam without a valid excuse, will **not** be admitted to the final examination and will receive the **NA** grade.

7. Make-up Policy

A make-up compensates for a *single* missing exam or missing laboratory. Students missing more than one exam or laboratory have to take multiple make-ups. Make-up exams can take place before or after the final exam; please follow the announcements on [ODTÜCLASS](#). The laboratory make-up is announced separately.

8. Previous Students of EE201 and EE213

- Students who have taken **EE213**, have a grade for it in their transcript, and are registered to the new EE201 will get their course grade from the EE201 theoretical part (100 %); previous laboratory grades will *not* be included.
- Students who **failed** EE213 and take only the EE213 laboratory will take their grades from the laboratory work only, as before.
- The new EE201 (which includes laboratory work) is opened only for new students who take this course for the first time.

9. Course Outline

- Basic Concepts** (6 hrs)
Introduction; lumped elements and lumped circuits; interconnection equations; branch relations; basic lumped elements; circuit analysis.
- Linear Time-Invariant Resistive Circuits** (10 hrs)
LTI resistive elements (resistors, dependent sources, ideal transformers); analysis methods; one-port circuits; two-ports.
- Nonlinear Resistive Circuits** (3 hrs)
Nonlinear resistive elements and circuits; load-line analysis; piecewise-linear resistive circuits.
- Operational Amplifier Circuits** (7 hrs)
Operational amplifiers; basic op-amp circuits; more realistic op-amp models; miscellaneous resistive op-amp circuits; op-amp with piecewise-linear elements.
- Dynamic Elements** (5 hrs)
Ramp, step and impulse functions; capacitors; inductors.
- First Order Circuits** (11 hrs)
First order linear differential equations with constant coefficients; simple LTI RC circuit; simple LTI RL circuit; analysis of miscellaneous LTI first order circuits; piecewise-linear first order circuits.

A more detailed course outline is available on [ODTÜCLASS](#).

10. Laboratory Outline

Exp.	Title	Exp.	Title
#1	Introduction to Voltage, Current and Resistance Measurements	#4	Resistive Circuits
#2	Signal Generators and Oscilloscopes	#5	Operational Amplifiers
#3	LTspice Workshop (<i>computer</i>)	#6	First Order Circuits